

Project Schedule: Forecast-Grounded Decision QA for Gas Pipeline Transient Operation

Timeframe: 8 weeks

Week 1 — Environment Setup

Day	Focus
Day 1	Read task sheet fully; clone and read both repos
Day 2	Get PipeFormer running locally on sample data
Day 3	Document PipeFormer's real input/output schema
Day 4	Get PipeClaw (backend + frontend) running locally
Day 5	Walk through one full PipeClaw question manually; find where it currently calls an LLM
Day 6	Inventory existing datasets (question_all, template_80/all, pipeclaw_v2, pipeformer_v4/v7)
Day 7	Confirm PipeFormer can be called programmatically from a script within PipeClaw's environment;

Weeks 2–3 — Task 1: Teacher Trace Dataset

Week	Focus	Output
Week 2	Build pressure/linepack/compressor rule library (priority rules only); build the parsing step (user_input → parsed_task)	Core *_rules.json files, working parser
Week 3	Build forecast + verification pipeline (parsed_task → PipeFormer → constraint_check); build evidence extraction + teacher_answer generation; run over a scoped seed subset; quality-check (schema + numerical + rule consistency, ~20% manual spot-check)	teacher_trace_train/valid/test.jsonl (scoped, quality-checked)

Milestone: A clean verified teacher trace dataset enough to distill from.

Weeks 4–5 — Task 2: Distillation & Fine-Tuning

Week	Focus	Output
Week 4	Select one candidate model (e.g. Qwen 7B); run answer-only SFT baseline; begin trace-level SFT	Baseline model, trace-distilled model (v1)
Week 5	Constraint-aware multi-task SFT pass; run 1–2 holdout splits (case holdout + one other) for a first generalization signal	Constraint-aware student model + initial generalization results

Milestone: A working distilled model that beats the answer-only baseline on the core tasks.

Weeks 6–8 — Task 3: Integration & Core Evaluation

Week	Focus	Output
Week 6	Integrate student model into PipeClaw; implement basic risk-tiered output gating	Working integrated system
Week 7	Run core evaluation: 3 comparison methods (LLM-only, answer-only SFT, constraint-aware student model) across 6 scenario types	Core evaluation results table
Week 8	Compile engineering + system metrics; produce key figures; write up findings and explicitly list what was deferred (full ablations, all 5 holdouts, teacher upper-bound comparison, real SCADA validation);	

Milestone: A working end-to-end demo with core metrics (accuracy, hallucination rate, latency, cost vs. baseline), plus a clear-eyed list of what a full-scope follow-on phase would still need to cover.